

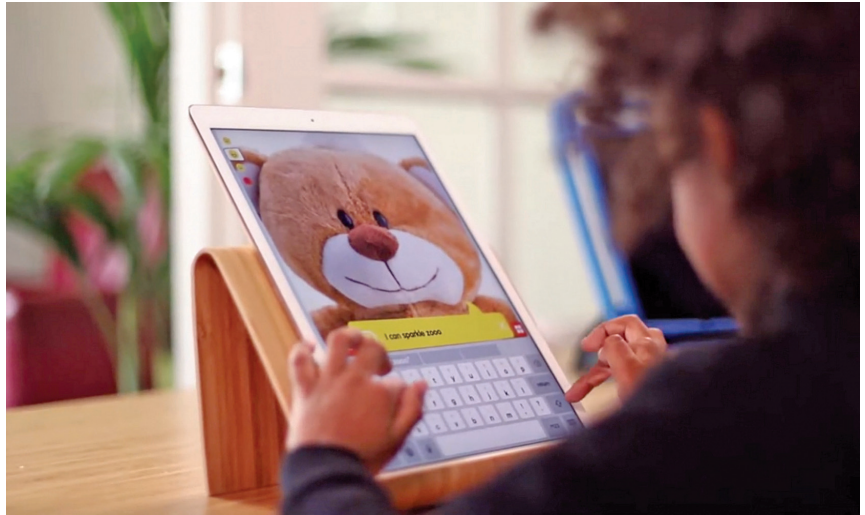
pushes doctors to defer the decision till the child goes to school and the symptoms become more pronounced. With advanced AI-based tools that can correlate and analyse behavioural patterns, neuroimaging, genetics, physiological responses, and other relevant parameters, doctors are now able to detect ASD as early as 18 months, leading to timely intervention and support. Advanced machine learning (ML) models help to analyse even subtle aspects such as eye gaze, speech, facial expressions, and motor skills.

Canvas Dx from Cognoa, is an FDA-authorised diagnostic solution that helps diagnose or rule out autism in children aged 1.5 to 6 years, within days. This smartphone application makes it possible to diagnose remotely, thereby supporting underserved regions where a qualified practitioner might not be available. Jaipur-based startup Aignosis offers similar AI-powered solutions for early screening, diagnosis, and detection of autism.

With data-driven AI tools like these, healthcare professionals can precisely assess an individual's specific needs, strengths, preferences, and challenges, and develop tailor-made solutions and treatment plans for each person. AI-powered systems also help to continuously track the individual's progress and responses, allowing therapy and support to be dynamically customised and improved.

Ranging from AI-powered augmentative and alternative communication (AAC) tools to social robots, and smart home apps and wearable devices, there are a whole lot of technologies out there that have the potential to make life better for individuals in the autistic spectrum. Let us look at some of them in this story.

However, do note that not all



iTherapy's InnerVoice uses AI innovatively to teach language, social interactions, and conversations in fun ways, without any stress (Source: iTherapy)

technologies suit everyone—and not everyone needs technology either! Therapists and counselors opt for no-tech, low-tech, or high-tech interventions based on an individual's capabilities and level of intervention required. It is therefore very important to consult your doctor before trying out any new technological aid.

Bolstering care and therapy

There is no such thing as stress-free caregiving. But AI-powered tools can share some of the caregivers' and therapists' responsibilities.

Foremost, AI can provide data-driven insights. Parents and other caregivers often find it difficult to give accurate information about an individual's condition or progress, due to the subjective nature of symptoms and emotional biases. However, AI tools automate data analysis and provide detailed progress reports, enabling the formulation of more targeted therapies. Some AI-powered remote monitoring tools not only help caregivers keep an eye on the individuals, but can also predict crises using behavioural analysis.

Another great advantage of

AI-driven apps and telehealth platforms for ASD management is the increased accessibility and democratisation of diagnosis and intervention, as even people in underserved or remote geographical areas get access to global best practices and specialised support from experts.

Here are some ways in which AI-powered tools support therapists, educators, and caregivers...

Behaviour analysis and tracking. AI-driven systems assist in data collection, behaviour tracking, and analysis, which can be used to personalise interventions. Modern AI-powered tools can record therapy sessions or day-to-day interactions of individuals with ASD and analyse the videos to track various aspects, such as eye movements, facial expressions, speech, motor skills, and engagement levels. Advanced tools can capture and analyse even subtle cues, such as fleeting eye contact, changes in tone, or gestures indicating frustration or engagement, which humans may miss. Therapeutic interventions and learning can be refined dynamically based on such real-time feedback and detailed data analysis by AI.